MILITARY SPECIFICATION

GYRO, STORED-ENERGY, FREE,

ROLL POSITION INDICATING

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the performance, design, test, manufacture and acceptance requirements for a roll position indicating free stored-energy gyro, herein referred to as the "gyro."

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-C-320	Chromium Plating (Electrodeposited).
QQ-N-290	Nickel Plating (Electrodeposited).

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1420

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QQ-P-35	Passivation Treatments for Corrosion-Resisting Steel.
QQ-P-416	Plating, Cadmium (Electrodeposited).
QQ-S-365	Silver Plating, Electrodeposited; General Requirements for.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-591	Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-665	Box, Paperboard, Metal Edged and Components.
PPP-B-676	Boxes, Setup.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing).
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	Preservation-Packaging, Methods of.
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-H-6088	Heat Treatment of Aluminum Alloys.
MIL-H-6875	Heat Treatment of Steels (Aerospace Practice, Process for).
MIL-H-7199	Heat Treatment of Wrought Copper-Beryllium Alloys, Process for (Copper Alloys: Numbers 170, 172 and 175).
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-T-10727	Tin Plating; Electrodeposited or Hot-Dipped, for Ferrous and Nonferrous Metals.
MIL-C-14550	Copper Plating (Electrodeposited).
MIL-C-26074	Coatings, Electroless Nickel, Requirements for.
MIL-G-45204	Gold Plating, Electrodeposited.
MIL-R-46085	Rhodium Plating, Electrodeposited.
MIL-H-81200	Heat Treatment of Titanium and Titanium Alloys.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

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MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-1523	Age Controls of Age-Sensitive Elastomeric Material.
MIL-STD-1695	Environments, Working, Minimum Standards for.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

1563560	Gyro, Stored-Energy, Free, Roll Position Indicating.

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

AS-2266	Guidance Section, Tactical, MARK 49 MOD 0 for Guided Missile AGM-45A/B-9 and MARK 49 MOD 1 for Guided Missile AGM-45A/B-9A.
WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AEROSPACE MATERIAL SPECIFICATION

AMS 7267	Rings, Sealing, Silicone Rubber.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 2 Pennsylvania Plaza, New York, NY 10001.)

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AEROSPACE STANDARDS

AS 568 Aerospace Size Standards for O-Rings.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 2 Pennsylvania Plaza, New York, NY 10001.)

UNIFORM FREIGHT CLASSIFICATION

Uniform Freight Classification 12 Ratings, Rules and Regulations.

(Application for copies should be made to the Uniform Classification Committee, 202 Union Station, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Item definition. The gyro is a roll position indicating free stored-energy gyro with 2 degrees of freedom (DOF). A two-wiper tapped and compensated potentiometer provides approximately the sine and cosine functions of outer gimbal rotation from the caged position. The gyro is provided with a caged and wound indicator and an inner gimbal limit indicator.

3.2 Characteristics.

3.2.1 Performance. The gyro shall perform as specified herein under the environmental conditions of 3.2.5.

3.2.1.1 Motor winding. A socket in the spring motor shaft shall receive a spring winding tool. The motor winding shaft shall ratchet during winding and be provided with a readily discernible full wind stop to prevent overwind damage. There shall be a visual means of verifying that the spring motor is fully wound.

3.2.1.2 Caging. The gyro shall be caged in all axes and shall meet the following caging requirements.

3.2.1.2.1 Caged restriction. With the gyro caged by the motor shaft and wound as in 3.2.1.1, no total angular play about the outer gimbal axis shall exceed 1.0 degree when the gimbal is rotated alternately clockwise (CW) and counterclockwise (CCW).

3.2.1.2.2 Caged indication. The gyro shall be delivered caged and with the motor fully wound. Under these conditions a resistance not exceeding 1.0 ohm, measured at 10 milliamperes direct current (mAdc), shall appear between terminals 3 and 4 (Drawing 1563560). A resistance of not less than 10 megohms, measured at 100 volts direct current (Vdc), shall appear between terminals 2 and 4 when the gyro is caged.

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3.2.1.3 Actuation. The actuation mechanism of the gyro shall have the following performance characteristics.

3.2.1.3.1 Actuator resistance. The direct current (dc) resistance between terminals 4 and 5 (Drawing 1563560) when measured using 10 mAdc maximum current with the gyro at 25 ±5 degrees Celsius (°C) shall be as follows:

(a) Prefired condition: 6.0 +1.3, -1.0 ohms.

(b) Postfired condition: Not less than 50 kilohms.

3.2.1.3.2 Actuator no-fire current. Application of 30 mAdc +0, -10 percent for 5 minutes minimum at +57 ±5°C between terminals 4 and 5 (Drawing 1563560) shall not fire the actuator.

3.2.1.3.3 Actuator firing current. A current of 300 mAdc +10, -0 percent applied between terminals 4 and 5 (Drawing 1563560) for 10 milliseconds (ms) shall cause the actuator to release the spring motor.

3.2.1.3.4 Uncaged indication. Within 100 ms maximum from onset of the current of 3.2.1.3.3, a resistance not exceeding 1.0 ohm, measured at 10 mAdc, shall appear between terminals 2 and 4 (Drawing 1563560). No caged indication shall recur with the gyro uncaged and operating. When the gyro is uncaged, a resistance not less than 10 megohms, measured at 100 Vdc, shall appear between terminals 3 and 4.

3.2.1.4 Runup time. The gyro shall exhibit a functional output within 100 ms maximum from the onset of the current of 3.2.1.3.3 when subjected to a roll rate of 360 ±10 degrees per second (deg/sec) in both CW and CCW directions when initiated with outer gimbal axis aligned with the horizontal ±10 degrees.

3.2.1.5 Running time. The gyro running time to gimbal lock shall be not less than 125 seconds and 100 seconds when subjected to a roll rate of 75 ±5 deg/sec and 260 ±10 deg/sec, respectively, in both CW and CCW directions when initiated with the outer gimbal axis aligned with the horizontal ±10 degrees at any temperature from +15 to +57°C. For temperatures from +15 to -43°C the running time requirement decreases linearly from 125 seconds to 100 seconds and from 100 seconds to 75 seconds, respectively.

3.2.1.6 Gimbal freedom.

3.2.1.6.1 Inner gimbal freedom. The rotor shall have 85 ±2 DOF of angular motion about the inner gimbal axis.

3.2.1.6.1.1 Inner gimbal lock indication. When the inner gimbal is within the limits of angular motion of 3.2.1.6.1, a resistance not less than 10 megohms, measured at 100 Vdc, shall appear between terminals 1 and 4

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(Drawing 1563560). Operating vibration and acceleration specified herein shall not cause the resistance to drop below 10 megohms at anytime, excluding 100 ms after receipt of firing current of 3.2.1.3.3. When the inner gimbal reaches either limit of angular motion of 3.2.1.6.1, a resistance not exceeding 1.0 ohm, measured at 10 mAdc, shall appear between terminals 1 and 4; this indicates gimbal lock.

3.2.1.6.2 Outer gimbal freedom. The outer gimbal shall have continuous 360 DOF of angular motion.

3.2.1.7 Pickoff potentiometer. The pickoff potentiometer shall have a continuous 360 degree linear resistance element excited across a diameter, the resistance element fixed to the gyro case with axis colinear with the outer gimbal axis; having orthogonal "sine" and "cosine" wipers, so oriented on the outer gimbal axle that the sine wiper is midway between resistance element excitation points when the gyro is caged; and having resistance element taps 45 degrees each way from each excitation point, with the taps adjacent to an excitation point jumpered together and the jumper connected to that excitation point through a fixed compensating resistance.

3.2.1.7.1 Excitation resistance. The resistance element shall be 10 kilohms ±5 percent total prior to tapping and padding and 2.50 kilohms ±5 percent as hypothetically measured between excitation points with both wipers and the four compensation taps unconnected. Each compensating resistance shall be 340 ohms ±1 percent as hypothetically measured across the isolated resistor. These requirements will be considered jointly met if resistance between terminals 6 and 7 (Drawing 1563560) is 1.69 kilohms ±5 percent by actual measurement with terminals 8 and 9 unconnected and the gyro at 25 ±5° C.

3.2.1.7.2 Resolution. One degree maximum rotation of the gyro about the outer gimbal axis at the caged position shall cause a change of 50 millivolts direct current (mVdc) minimum in output voltage from either wiper into a 105 kilohm ±1 percent resistance to ground when excitation across the diameter is 22 ±2 Vdc.

3.2.1.7.3 Wiper contact. With excitation and wiper electrical loading as in 3.2.1.7.2, output noise from either wiper in the frequency band 0 to 50 hertz (Hz) shall not exceed 100 millivolts (mV) peak-to-peak at any time during tests for compliance with the drift requirement of 3.2.1.8.1 or 3.2.1.8.2.

3.2.1.7.4 Sine and cosine approximation. When wiper outputs versus outer gimbal angle are recorded, at no generated outer gimbal angle shall the arc sine or arc cosine (as appropriate) output differ by more than 10 degrees from the generated angle, or by more than 2.0 degrees with the gyro caged and wound. The wiper position when the gyro is caged and wound shall be such that

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when the gyro is uncaged and rotated CW (Figure 1) about the outer gimbal axis an angle ϕ, the wiper outputs shall be sine ϕ on terminal 8 (Drawing 1563560) and - cosine ϕ on terminal 9.

3.2.1.7.5 Uncaging offset. With excitation and wiper loading as in 3.2.1.7.2, the arc sine of the sine output shall not be changed more than 2 degrees by actuating the stationary gyro.

3.2.1.8 Drift.

3.2.1.8.1 Fast roll. When tested as specified herein, the outer gimbal drift measured after the specified running time (3.2.1.5) after actuation shall not exceed 20 degrees.

3.2.1.8.2 Slow roll. When tested as specified herein, the outer gimbal drift measured after the specified running time (3.2.1.5) after actuation shall not exceed 20 degrees.

3.2.1.9 Electrical characteristics.

3.2.1.9.1 Dielectric withstanding voltage. With terminals 4 and 5 jumpered (Drawing 1563560) but no other connection to any but the terminal under test, the gyro shall without disruptive discharge or impairment of subsequent function withstand 200 volts root-mean-square (Vrms), 60 Hz, applied for 60 seconds minimum in preproduction or lot testing or 1 to 3 seconds in quality conformance testing, between each gyro terminal and the gyro case when tested as specified herein.

3.2.1.9.2 Insulation resistance. With terminals 4 and 5 jumpered (Drawing 1563560), insulation resistance between electrically isolated interior circuits and between each such circuit and the gyro case shall be 10 megohms minimum when measured as specified herein.

3.2.1.9.3 Actuator insulation resistance. After firing as in 3.2.1.3.3, the resistance between terminal 5 (Drawing 1563560) and the gyro case shall be 50 kilohms minimum.

3.2.1.10 Terminal strength. Each gyro terminal shall withstand a lateral pull on the outer end, and an outward radial pull, each of 2 pounds minimum applied for 30 seconds minimum, without loosening or seal breakage.

3.2.2 Physical characteristics.

3.2.2.1 Weight. The gyro shall not weigh more than 18 ounces.

3.2.2.2 Dimensions. Gyro outline and mounting dimensions shall not exceed the limits specified in Drawing 1563560.

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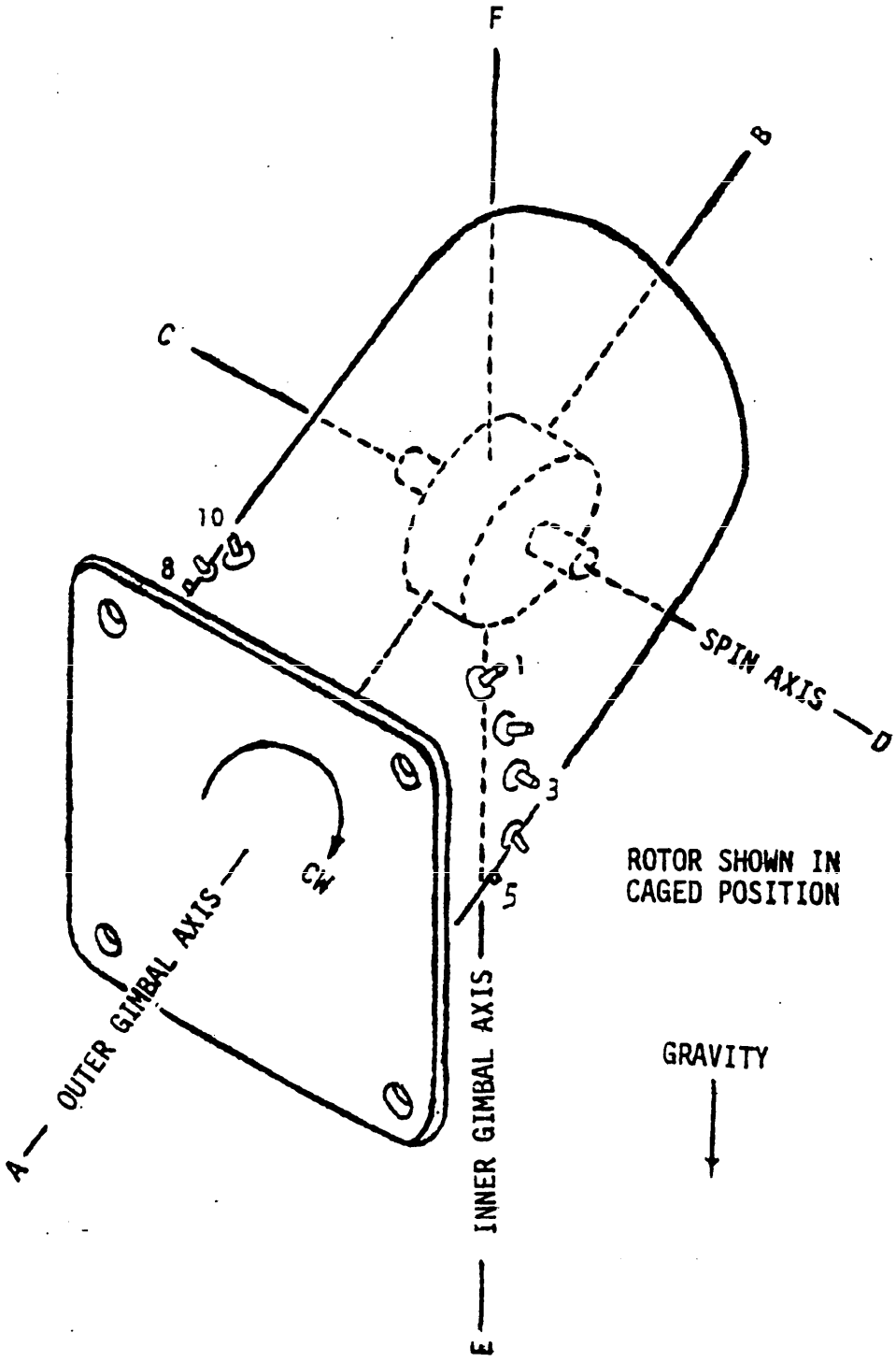


FIGURE 1. Gyro reference axes.

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3.2.2.3 Seal.

3.2.2.3.1 Cover. The gyro cover shall be primary-sealed to the gyro base. The cover shall be readily removable from the base after the primary seal is severed. Cover and base design shall provide a gland and mating sealing surface for installation of an O-ring of an AS 568 standard size, of material in accordance with AMS 7267, as a secondary seal. Cover design shall provide for flushing the gyro with dry air or nitrogen when the cover is rejoined to the base with the secondary seal installed.

3.2.2.3.2 Filling and leakage. The interior of the gyro as delivered shall be filled with a mixture of 95 ±1 percent helium and 5 ±1 percent carbon dioxide by volume. The mixture shall be free of particles larger than 0.1 micron in maximum dimension, and shall contain no more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling pressure shall be 14.7 ±0.5 pounds per square inch (psi) absolute. Leakage detected during fine leak test shall not exceed 10^{-7} atmospheric cubic centimeter per second at 1.0 atmosphere minimum pressure differential when tested as specified herein.

3.2.2.4 Life. Gyro design shall provide for at least 12 actuations of 3.2.1.3.3 for gyro uncage and rundown operations.

3.2.3 Reliability. The design reliability of the item shall be established by the selection and inspection of materials, parts, and processes used in fabrication and shall be 0.999 which shall be demonstrated by prediction. The performance of the item shall be sufficiently reliable to successfully meet all requirements of preproduction inspection with production reliability demonstration to be 0.843 at 80 percent confidence. This shall be accomplished by completing the preproduction inspection of 4.1.2.1 with no failures. Subsequent to preproduction inspection, reliability shall be maintained by control of materials, parts, and processes and verified by the quality conformance inspection of 4.2.

3.2.4 Maintainability. The gyro shall be designed so as to require no periodic maintenance.

3.2.5 Environmental conditions. The gyro shall meet the requirements specified in 3.2.1 after exposure to the nonoperating environments, and during and after exposure to the operating environments specified herein.

3.2.5.1 Nonoperating environments.

3.2.5.1.1 Temperature cycle. The gyro shall perform as specified in 3.2.1 after being subjected to temperature cycling at +57 ±5°C and -43 ±5°C in the manner specified herein.

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3.2.5.1.2 Sine vibration. The gyro shall perform as specifed in 3.2.1 after being subjected to a sine vibration frequency at a logarithmic rate from 5 to 500 to 5 Hz in 15 minutes in the manner specified herein with the gyro at a temperature of -43 and +57°C, respectively.

3.2.5.1.3 Random vibration. The gyro shall perform as specified in 3.2.1 after being subjected to random vibration of 20 to 2,000 Hz in the manner specified herein with the gyro at a temperature of -43 and +57°C, respectively.

3.2.5.1.4 Temperature shock and altitude. The gyro shall perform as specified in 3.2.1 after being subjected to six temperature shock cycles ranging from +71 ±5°C at ambient barometric pressure to -43 ±5°C at a 50,000-foot barometric altitude in the manner specified herein.

3.2.5.1.5 Handling shock. The gyro shall perform as specified in 3.2.1 after being subjected to a total of 36 handling shock cycles in the manner specified herein with the gyro at a temperature of -43 and +57°C, respectively.

3.2.5.1.6 Humidity. The gyro shall perform as specified in 3.2.1 after being subjected to the humidity environment test specified herein.

3.2.5.2 Operating environments.

3.2.5.2.1 Temperatures. The gyro shall operate as specified in 3.2.1 when actuated at any gyro temperature from -43 to +57°C.

3.2.5.2.2 Acceleration. The gyro shall operate as specified in 3.2.1 when exposed to 25 ±3 gravity units (g) acceleration applied along the A-B axis only (Figure 1), in the direction A to B, for 5 seconds minimum starting at the onset of the voltage of 3.2.1.3.3.

3.2.5.3 Storage life. The gyro shall be designed to withstand 5 years of sheltered storage and then meet the requirements specified herein.

3.2.6 Transportability. Transportation requirements shall be in accordance with the Uniform Freight Classification Rules which permit shipment under the minimum requirements for a device of this type.

3.3 Design and construction. Gyro design and construction shall conform to Drawing 1563560 and comply with the following.

3.3.1 Materials, processes and parts. Materials, processes and parts to which particular MIL-STD-454 requirements pertain shall, wherever possible, comply with those requirements.

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3.3.1.1 Materials.

3.3.1.1.1 Fungus resistance. Only materials of MIL-STD-454, Requirement 4, Group I (fungus-inert in all states and grades) shall be exposed on the gyro exterior. Other materials are permissible in the hermetically sealed interior.

3.3.1.1.2 Age-sensitive elastomers. The gyro shall incorporate no elastomers of type to which MIL-STD-1523 applies age controls.

3.3.1.1.3 Lubricants. Gyro lubricants shall be consistent with storage life requirements of 3.2.3.3.

3.3.1.1.4 Adhesives. Any adhesive within or on the exterior of the gyro shall comply with MIL-STD-454, Requirement 23.

3.3.1.1.5 Dissimilar metals. Dissimilar metals shall not be used in direct contact unless protected against eletrolytic corrosion. Selection and protection of dissimilar metal combinations shall meet the Class 2 provisions for gyro exterior materials, and the Class 3 provisions for gyro interior materials of MIL-STD-454, Requirement 16.

3.3.1.1.6 Magnesium. The gyro shall incorporate no pure magnesium nor alloy in which magnesium is the major constituent.

3.3.1.1.7 Potentiometer wire size. Winding wire in the gyro pickoff shall have a diameter no less than 0.0010 inch nominal or 0.0009 inch absolute minimum.

3.3.1.2 Processes.

3.3.1.2.1 Brazing. Any brazing in gyro fabrication shall comply with MIL-STD-454, Requirement 59.

3.3.1.2.2 Welding. Any structural welding in gyro fabrication shall comply with MIL-STD-454, Requirement 13.

3.3.1.2.3 Heat treatment. Any heat treatment in gyro fabrication shall comply with MIL-H-6088 for aluminum alloys; MIL-H-6875 for steels; MIL-H-7199 for beryllium copper; or MIL-H-81200 for titanium.

3.3.1.2.4 Plating. Any plating in gyro fabrication shall be by processes minimizing long-term flaking and particulate contamination. Plating shall comply with QQ-C-320, Class 2, Type II, for chromium; QQ-N-290, Class 1, or MIL-C-26074, Class 1 or 2, for nickel; QQ-P-416, Class 2, Type II, for cadmium, with no cadmium to be used in contact with titanium or on parts on

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moving contact; QQ-S-365, Type II, Grade B, for silver; MIL-G-45204, Type II, for gold; MIL-R-46085, Type I, Class 2, for rhodium; MIL-C-14550, for copper; or MIL-T-10727, for tin. Zinc plating shall not be used.

3.3.1.2.5 Passivation. Corrosion and heat resistant steels in the gyro shall be passivated in accordance with QQ-P-35.

3.3.1.2.6 Aluminum surfaces. Aluminum or aluminum-alloy parts in the gyro shall be anodized in accordance with MIL-A-8625 or chemically coated in accordance with MIL-C-5541, with subsequent painting permissible.

3.3.1.2.7 Soldering. Soldering of electrical conductors in the gyro shall comply with WS-6536 or an equivalent specification subject to approval by the procuring activity.

3.3.1.3 Parts.

3.3.1.3.1 Springs. Springs in the gyro shall comply with MIL-STD-454, Requirement 41.

3.3.1.3.2 Bearings. Bearings in the gyro shall comply with MIL-STD-454, Requirement 6.

3.3.1.3.3 Castings. Castings in the gyro shall comply with MIL-STD-454, Requirement 21.

3.3.1.3.4 Resistors. Electrical resistors in the gyro shall comply with MIL-STD-454, Requirement 33, with Established Reliability (ER) types to be used where available.

3.3.2 Electromagnetic radiation. The gyro shall not conduct, radiate, nor be susceptible to electromagnetic radiation as specified in AS-2266.

3.3.3 Identification and marking. The gyro shall be identified in accordance with MIL-STD-130 by a marking that will not deteriorate under environmental conditions of 3.2.5. At minimum, the marking shall include Part Number 1563560 and revision letter, a unique serial number, manufacturer's name or symbol and Federal Supplier Code, if any, manufacturer's part number, and date of manufacture. Terminals shall be labeled 1 through 9.

3.3.4 Interchangeability. Gyro interchangeability shall be in accordance with MIL-STD-454, Requirement 7.

3.3.5 Standards of manufacture.

3.3.5.1 Cleanliness. All critical gyro assembly operations shall take place in a clean environment meeting Class 100 requirements of MIL-STD-1695. Before final assembly, all components and subassemblies shall be cleaned in

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accordance with a procedure prepared by the contractor, who shall have available either a copy of these procedures or be able to certify that MIL-STD-1695 Class 100 requirements are met.

3.4 First article. Unless otherwise specified in the contract (see 6.2), the contractor shall furnish 14 gyros for first article inspection and approval (see 4.1.2.1 and 6.2). The first article sample shall be manufactured using the same design, materials, and processes proposed for production, and be subjected to quality conformance inspection. Any production prior to approval of the first article sample shall be at the contractor's risk.

3.5 Gyro workmanship. Gyro workmanship shall comply with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General. This section provides for verification of compliance with the requirements of Sections 3 and 5 by examination and test of the gyro, and by examination of manufacturing data, certifications, and gyro preparation for delivery.

4.1.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.1.2.1).

(b) Lot inspection (see 4.2.2).

(c) Quality conformance inspection (see 4.2).

4.1.2 Special tests and examinations.

4.1.2.1 Preproduction inspection. Seven gyros shall be randomly selected from the first article sample of 3.4 and be subjected to preproduction inspections specified in Table I. Unless otherwise specified in the contract (see 6.2), any design change may require preproduction inspection to be repeated.

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TABLE I. Preproduction inspection.

Inspection requirement	Conditions and reference paragraph	Requirement paragraph	Inspection paragraph	Spec. Nos. 1/
Table II, Groups 2 and 3	Ambient conditions 2/ (4.3)	Table II	Table II	All
Dimensions		3.2.2.2	4.4.1	All
Temperature cycling and vibration:	(Nonoperating preconditioning)			
Temperature cycle		3.2.5.1.1	4.4.9.1.1	All
Sine vibration	+57°C, Axes A-B, C-D, E-F	3.2.5.1.2	4.4.9.1.2	All
Sine vibration	-43°C, same as above			All
Random vibration	+57°C, same as above	3.2.5.1.3	4.4.9.1.3	All
Random vibration	-43°C, same as above			All
Temperature shock and altitude	(Nonoperating preconditioning)	3.2.5.1.4	4.4.9.1.4	All
Handling shock:	(Nonoperating preconditioning)	3.2.5.1.5	4.4.9.1.5	
Test 1	+57°C, Axes A-B, C-D, E-F			All
Test 2	-43°C, same as above			All
Humidity	(Nonoperating preconditioning)	3.2.5.1.6	4.4.9.1.6	5, 6, 7, and 8
Table II, Group 2	Ambient conditions 2/ (4.3.1.1)	- - - -	- - - -	All
Terminal strength		3.2.1.10	4.4.7.3	All
Leakage	Fine (4.4.8.1) Gross (4.4.8.2)	3.2.2.3.2	4.4.8	All
Cover removal		3.2.2.3.1	4.3.1	All
Caged restriction		3.2.1.2.1	4.4.2.1	All
Cover replacement		3.2.2.3.1	4.3.1	All

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TABLE I. Preproduction inspection. - Continued

Inspection requirement	Conditions and reference paragraph	Requirement paragraph	Inspection paragraph	Spec. Nos. 1/
Actuation 1:	-43°C	3.2.1.3.3	4.4.3	All 3/
1. Uncaged indication		3.2.1.3.4	4.4.3.4	
2. Outer gimbal freedom		3.2.1.6.2	4.4.4.3	
3. Resolution		3.2.1.7.2	4.4.5.2	
4. Sine and cosine approx.		3.2.1.7.4	4.4.5.4	
5. Uncaging offset		3.2.1.7.5	4.4.5.5	
6. Inner gimbal lock indication		3.2.1.6.1.1	4.4.4.2	
7. Actuator resistance (post-fire)		3.2.1.3.1	4.4.3.1	
8. Actuator insulation resistance		3.2.1.9.3	4.4.7.4	
9. Replace actuator (repeat after each subsequent activation)	Ambient condition	- - -	4.3.1	All 3/
Actuation 2: (specimens 1, 3, 5, 7) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	-43°C, Fast roll, CCW (4.4.6.1)	3.2.1.3.3	4.4.3	1, 3, 5, 7
1. Runup time		3.2.1.4	4.4.3.5	
2. Wiper contact		3.2.1.7.3	4.4.5.3	
3. Fast roll drift		3.2.1.8.1	4.4.6.1	
4. Running time		3.2.1.5	4.4.3.6	

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TABLE I. Preproduction inspection. - Continued

Inspection requirement	Conditions and reference paragraph	Requirement paragraph	Inspection paragraph	Spec. No. 1/
Actuation 2: (specimens 2, 4, 6) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	-43°C, Slow roll, CCW (4.4.6.2)	3.2.1.3.3	4.4.3	2, 4, 6
1. Runup time		3.2.1.4	4.4.3.5	
2. Wiper contact		3.2.1.7.3	4.4.5.3	
3. Slow roll drift		3.2.1.8.2	4.4.6.2	
4. Running time		3.2.1.5	4.4.3.6	
Actuation 3: (specimens 1, 3, 5, 7) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	-43°C, Fast roll, CW	3.2.1.3.3	4.4.3	1, 3, 5, 7
1. Runup time		3.2.1.4	4.4.3.5	
2. Wiper contact		3.2.1.7.3	4.4.5.3	
3. Fast roll drift		3.2.1.8.1	4.4.6.1	
4. Running time		3.2.1.5	4.4.3.6	
Actuation 3: (specimens 2, 4, 6) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	-43°C, Slow roll, CW	3.2.1.3.3	4.4.3	2, 4, 6
1. Runup time		3.2.1.4	4.4.3.5	
2. Wiper contact		3.2.1.7.3	4.4.5.3	
3. Slow roll drift		3.2.1.8.2	4.4.6.2	
4. Running time		3.2.1.5	4.4.3.6	

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TABLE I. Preproduction inspection - Continued

Inspection requirement	Conditions and reference paragraph	Requirement paragraph	Inspection paragraph	Spec. No. 1/
Actuation 4: (specimens 1, 3, 5, 7) Repeat items 1, 2, 5, 7 and 8 under actuation 1.	-43°C, Slow roll, CCW	3.2.1.3.3	4.4.3	1, 3, 5, 7
1. Runup time 2. Wiper contact 3. Slow roll drift 4. Running time		3.2.1.4 3.2.1.7.3 3.2.1.8.2 3.2.1.5	4.4.3.5 4.4.5.3 4.4.6.2 4.4.3.6	
Actuation 4: (specimens 2, 4, 6) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	-43°C, Fast roll, CW	3.2.1.3.3	4.4.3	2, 4, 6
1. Runup time 2. Wiper contact 3. Fast roll drift 4. Running time		3.2.1.4 3.2.1.7.3 3.2.1.8.1 3.2.1.5	4.4.3.5 4.4.5.3 4.4.6.1 4.4.3.6	
Actuation 5: (all specimens) Repeat items 1 thru 8 under actuation 1.	-43°C	3.2.1.3.3	4.4.3	All
Actuation 6: (specimens 1, 3, 5, 7) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	-43°C, Slow roll, CW	3.2.1.3.3	4.4.3	1, 3, 5, 7
1. Runup time 2. Wiper contact 3. Slow roll drift 4. Running time		3.2.1.4 3.2.1.7.3 3.2.1.8.2 3.2.1.5	4.4.3.5 4.4.5.3 4.4.6.2 4.4.3.6	

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TABLE I. Preproduction inspection. - Continued

Inspection requirement	Conditions and reference paragraphs	Requirement paragraph	Inspection paragraph	Spec. Nos. 1/
Actuation 6: (specimens 2, 4, 6) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	-43°C, Fast roll, CCW	3.2.1.3.3	4.4.3	2, 4, 6
1. Runup time		3.2.1.4	4.4.3.5	
2. Wiper contact		3.2.1.7.3	4.4.5.3	
3. Fast roll drift		3.2.1.8.1	4.4.6.1	
4. Running time		3.2.1.5	4.4.3.6	
Actuation 7: (specimens 1, 2, 3, 4) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	+57°C, Fast roll, CCW	3.2.1.3.3	4.4.3	1, 2, 3, 4
1. Runup time		3.2.1.4	4.4.3.5	
2. Wiper contact		3.2.1.7.3	4.4.5.3	
3. Fast roll drift		3.2.1.8.1	4.4.6.1	
4. Running time		3.2.1.5	4.4.3.6	
Actuation 7: (specimens 5, 6, 7) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	+57°C, Slow roll, CCW	3.2.1.3.3	4.4.3	5, 6, 7
1. Runup time		3.2.1.4	4.4.3.5	
2. Wiper contact		3.2.1.7.3	4.4.5.3	
3. Slow roll drift		3.2.1.8.2	4.4.6.2	
4. Running time		3.2.1.5	4.4.3.6	
Actuation 8: (all specimens) Repeat items 1 thru 8 under actuation 1.	+57°C	3.2.1.3.3	4.4.3	All

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TABLE I. Preproduction inspection. - Continued

Inspection requirement	Conditions and reference paragraph	Requirement paragraph	Inspection paragraph	Spec. Nos. 1/
Actuation 9: (specimens 1, 2, 6, 7) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	+57°C, Fast roll, CW	3.2.1.3.3	4.4.3	1, 2, 6, 7
1. Runup time		3.2.1.4	4.4.3.5	
2. Wiper contact		3.2.1.7.3	4.4.5.3	
3. Fast roll drift		3.2.1.8.1	4.4.6.1	
4. Running time		3.2.1.5	4.4.3.6	
Actuation 9: (specimens 3, 4, 5) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	+57°C, Slow roll, CW	3.2.1.3.3	4.4.3	3, 4, 5
1. Runup time		3.2.1.4	4.4.3.5	
2. Wiper contact		3.2.1.7.3	4.4.5.3	
3. Slow roll drift		3.2.1.8.2	4.4.6.2	
4. Running time		3.2.1.5	4.4.3.6	
Actuation 10: (specimens 1, 2, 3, 4) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	+57°C, Slow roll, CCW	3.2.1.3.3	4.4.3	1, 2, 3, 4
1. Runup time		3.2.1.4	4.4.3.5	
2. Wiper contact		3.2.1.7.3	4.4.5.3	
3. Slow roll drift		3.2.1.8.2	4.4.6.2	
4. Running time		3.2.1.5	4.4.3.6	

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TABLE I. Preproduction inspection. - Continued

Inspection requirement	Conditions and reference paragraph	Requirement paragraph	Inspection paragraph	Spec. Nos. 1/
Actuation 10: (specimens 5, 6, 7) Repeat items 1, 2, 6, 7 and 8 under actuation 1.	+57°C, Fast roll, CCW	3.2.1.3.3	4.4.3	5, 6, 7
1. Runup time		3.2.1.4	4.4.3.5	
2. Wiper contact		3.2.1.7.3	4.4.5.3	
3. Fast roll drift		3.2.1.8.1	4.4.6.1	
4. Running time		3.2.1.5	4.4.3.6	
Actuation 11: (all specimens) Repeat items 1, 5, 6, 7 and 8 under actuation 1.	Acceleration - ambient	3.2.1.3.3 3.2.5.2.2	4.4.3 4.4.9.2.2	All
Actuation 12: (all specimens) Repeat items 1 thru 8 under actuation 1.	+57°C	3.2.1.3.3	4.4.3	All

NOTES:

1/ Specimen numbers inspected. "All" refers to Specimens 1, 2, 3, 4, 5, 6, 7, and 8.

2/ Except gyro temperature of +57°C minimum for actuator no-fire current test.

3/ Specimen 8 set aside here as reserve against accidental loss of any other specimen. To substitute for a lost specimen, No. 8 shall first undergo stresses accumulated by lost specimen to point of lost.

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4.1.2.1.1 Action in event of preproduction inspection failure. Failure of one or more gyros of the preproduction sample to meet all requirements of the preproduction inspection (Table I) shall reject the preproduction sample. The contractor shall then take action acceptable to the procuring activity to correct the cause(s) of failure. No production lot of gyros will be considered for acceptance unless the production lot reflects the corrective action or until a new preproduction sample reflecting the same action passes preproduction inspection.

4.2 Quality conformance inspection. Every gyro shall be subjected to the quality conformance inspection of Table II, with procedural sequence as shown for Group 1 characteristics and sequence optional for Groups 2 and 3.

4.2.1 Action in event of quality conformance inspection failure. Failure of a production gyro to meet all Table II, Group 1 requirements, or all Group 2 requirements for which the gyro is individually inspected, shall reject that gyro only. With Procuring Activity approval, a gyro rejected only for failure to meet individual requirements may be reworked once only and resubmitted for acceptance. Failure of a production lot of gyros to meet all Group 2 requirements for which the lot is inspected collectively, by examination of manufacturing data or certifications, shall reject the lot. With Procuring Activity approval, a lot so rejected may be reworked once only and resubmitted for acceptance. Failure of packaging/packing of a production lot of gyros to meet all Table II, Group 3 requirements shall reject the lot. With Procuring Activity approval, a lot so rejected may be repackaged/repacked once only and resubmitted for acceptance, subject only to Group 3 reinspection and reinspection for the Group 1 requirement of presence of caged and wound indication.

4.2.2 Lot inspection. Lot sample gyros shall be randomly selected from each lot offered for acceptance in accordance with MIL-STD-105 and shall be tested in the sequence of Table III.

4.2.2.1 Inspection lot definition. An inspection lot shall be as defined in MIL-STD-105.

4.2.2.2 Lot sampling. After being subjected to the individual tests, a lot sample shall be drawn randomly in accordance with MIL-STD-105, General Inspection Level II, Single Sampling, Normal Inspection, with an Acceptable Quality Level (AQL) of 1 percent.

4.2.2.3 Acceptance criteria. Failure of the sample to meet the acceptance criteria of MIL-STD-105 shall cause rejection of the lot.

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TABLE II. Gyro quality conformance inspection.

Inspection or test	Requirement paragraph	Inspection paragraph
Group 1 1/		
Manual Actuation 1 (Ambient, CW, Slow roll):		
1. Uncaged indication	3.2.1.3.4	4.4.3.4
2. Running time	3.2.1.5	4.4.3.6
3. Inner gimbal lock	3.2.1.6.1.1	4.4.4.2
4. Slow roll drift	3.2.1.8.2	4.4.6.2
Manual Actuation 2 (Ambient, CCW, Fast roll):		
1. Uncaged indication	3.2.1.3.4	4.4.3.4
2. Running time	3.2.1.5	4.4.3.6
3. Inner gimbal lock	3.2.1.6.1.1	4.4.4.2
4. Fast roll drift	3.2.1.8.1	4.4.6.1
Group 2 2/		
Actuator no-fire current	3.2.1.3.2	4.4.3.2
Actuator resistance (pre-fired)	3.2.1.3.1	4.4.3.1
Caged indication (present)	3.2.1.2.2	4.4.2.2
Uncaged indication (absent)	3.2.1.3.4	4.4.3.4
Inner gimbal lock indication (absent)	3.2.1.6.1.1	4.4.4.2
Pickoff excitation resistance	3.2.1.7.1	4.4.5.1
Dielectric withstanding voltage	3.2.1.9.1	4.4.7.1
Insulation resistance	3.2.1.9.2	4.4.7.2
Group 3 3/		
Weight and dimensions	3.2.2.1	4.4.1(d)
Materials, processes and parts	3.3.1	4.4.1(g)
Identification and marking	3.3.3	4.4.1(h)
Workmanship (exterior)	3.5	4.4.1(i)
Cleanliness	3.3.8.1	4.4.10(c)
Group 4 2/		
Gross leak test	3.2.2.3.2	4.4.8.2

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TABLE II. Gyro quality conformance inspection. - Continued

Inspection or test	Requirement paragraph	Inspection paragraph
Group 5 4/ Preparation for delivery	5	4.4.1

NOTES:

1/ Tests to be conducted prior to the primary seal operation. Verification of tests by examination of manufacturing data.

2/ Verification by gyro test and/or measurement, at ambient conditions except gyro temperature +57°C for actuator no-fire current test.

3/ Verification by visual examination of gyro and/or examination of manufacturing data and manufacturer's certifications, except weight measures.

4/ Verification by visual examination of gyro packaging and packing.

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TABLE III. Lot inspection.

Inspection or test	Condition and reference paragraph	Requirement paragraph	Inspection paragraph
Table II, Groups 2, 3, and 4 inspections	Ambient temp. conditions (see 4.3)	Table II	Table II
Dimensions		3.2.2.2	4.4.1
Temp. cycling and vibration temp. cycle:	Nonoperating, preconditioning	3.2.5.1.1	4.4.9.1.1
1. Random vibration	+57°C, Axes A-B, C-D, E-F	- - -	- - -
2. Random vibration	-43°C, Axes A-B, C-D, E-F	- - -	- - -
Handling shock:	Nonoperating, preconditioning	3.2.5.1.5	4.4.9.1.5
1. Test 1	+57°C, Axes A-B, C-D, E-F	- - -	- - -
2. Test 2	-43°C, Axes A-B, C-D, E-F	- - -	- - -
Table II, Group 2	Ambient temp. conditions (see 4.3)	Table II	Table II
Terminal strength		3.2.1.10	4.4.1
Leakage	Fine (4.4.8.1) Gross (4.4.8.2)	3.2.2.3.2	4.4.8
Actuation 1:	Ambient (see 4.3)	3.2.1.3.3	4.4.3
1. Uncaged indication		3.2.1.3.4	4.4.3.4
2. Outer gimbal freedom		3.2.1.6.2	4.4.4.3
3. Resolution		3.2.1.7.2	4.4.5.2
4. Sine and cosine approximation		3.2.1.7.4	4.4.5.4
5. Uncaging offset		3.2.1.7.5	4.4.5.5
6. Inner gimbal lock indication		3.2.1.6.1.1	4.4.4.2
7. Actuator resistance (post-fire)		3.2.1.3.1	4.4.3.1
8. Actuator insulation resistance		3.2.1.9.3	4.4.7.4
9. Replace actuator (repeat after each subsequent activation)		- - -	4.3

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TABLE III. Lot inspection. - Continued

Inspection or test	Condition and reference paragraph	Requirement paragraph	Inspection paragraph
Actuation 2:	-43°C, Fast roll CW (see 4.4.6.1)	3.2.1.3.3	4.4.3
1. Uncaged indication		3.2.1.3.4	4.4.3.4
2. Runup time		3.2.1.4	4.4.3.5
3. Outer gimbal freedom		3.2.1.6.2	4.4.4.3
4. Wiper contact		3.2.1.7.3	4.4.5.3
5. Running time		3.2.1.5	4.4.3.6
6. Inner gimbal lock indication		3.2.1.6.1.1	4.4.4.2
7. Actuator resistance (post-fire)		3.2.1.3.1	4.4.3.1
8. Actuator insulation resistance		3.2.1.9.3	4.4.7.4
9. Fast roll drift		3.2.1.8.1	4.4.6.1
Actuation 3:	+57°C, Slow roll, CCW (see 4.4.6.2)	3.2.1.3.3	4.4.3
Repeat items 1 thru 8 under actuation 2.			
1. Slow roll drift		3.2.1.8.2	4.4.6.2
Actuation 4:	Ambient, Fast roll, CCW (see 4.4.6.1)	3.2.1.3.3	4.4.3
Repeat items 1 thru 9 under actuation 2.			
Actuation 5:	Ambient, Slow roll, CW (see 4.4.6.2)	3.2.1.3.3	4.4.3
Repeat items 1 thru 8 under actuation 2.			
1. Slow roll drift		3.2.1.8.2	4.4.6.2
Actuation 6: Repeat items 1 thru 9 under actuation 1.	Ambient (see 4.3)	3.2.1.3.3	4.4.3

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4.3 Inspection, equipment, facilities, and conditions. Inspection equipment and facilities shall be adequate in quality and quantity to permit performance of all specified examinations and tests. Environmental conditions not otherwise specified shall be within the following limits during inspections.

(a) Air temperature: 15 to 35°C.

(b) Relative humidity: 10 to 95 percent.

(c) Barometric pressure: Ambient, at ground elevation to 7,000 feet maximum.

4.3.1 Clean room. Removal of the gyro cover, gyro operation with cover removed (as for measurement of runup time), actuator replacement, motor rewinding, and cover replacement, shall take place only in a clean environment meeting Class 100 requirements of MIL-STD-1695.

4.3.2 Test equipment accuracy and calibration. Test equipment aggregate error (from inaccuracy, limitation of resolution, and normal error of perception) shall not exceed one-tenth of the tolerance upon the parameter tested; or in the instance of a single-limit tolerance, the error shall be applied to the observation in such direction as to assure that the parameter satisfies the limit. Test equipment calibration shall be traceable to the National Bureau of Standards.

4.3.3 Tolerances on specific test conditions. Unless otherwise specified herein, the maximum allowable tolerances on specified test conditions, exclusive of equipment accuracy, shall be as follows:

(a)	Temperature:	±2°C.
(b)	Altitude:	±5 percent of pressure value.
(c)	Relative humidity:	+5, -0 percent.
(d)	Vibration acceleration:	
	(1) Sinusoidal:	±10 percent of peak.
	(2) Random:	1.5 decibel (dB) maximum of power spectral density or +40, -30 percent rootmean-square (rms).
(e)	Vibration frequency:	±2 percent.
(f)	Static loads:	+2, -0 percent.

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(g)	Shock loads:	+0, -10 percent.
(h)	Shock duration:	±10 percent.
(i)	Voltage, alternating current (ac) (rms) or direct current (dc):	±1 percent.
(j)	Voltage, ac (peak-to-peak):	±5 percent.
(k)	Current, dc:	±1 percent.
(l)	Frequency:	±1 percent.
(m)	Time:	±1 percent.
(n)	Resistance (dc):	±1 percent.

4.3.4 Temperature stabilization. When tests are required at low or high temperatures, the gyro shall be conditioned at the specified temperature for not less than 1 hour prior to testing. If the gyro is removed from the temperature chamber for performance of the tests, all testing shall be completed within 20 minutes after removal. If the tests are not completed within the 20 minute period, the gyro shall be returned to the temperature chamber for 1 hour before continuing the tests.

4.4 Test and inspection methods. The following methods shall be applied in inspections at the times and under the circumstances indicated in Tables I, II, and III.

4.4.1 Visual and mechanical inspection. Visually and with mechanical gagings as necessary, the gyro and its preparation for delivery shall be examined for compliance with the following:

	Item	Requirement paragraph
(a)	Motor winding:	3.2.1.1
(b)	Caging restrictions:	3.2.1.2.1, 4.4.2.1
(c)	Terminal strength:	3.2.1.10
(d)	Weight and dimensions:	3.2.2.1, 3.2.2.2
(e)	Terminal and cover seal:	3.2.2.3.1, 3.2.2.3.2
(f)	Life:	3.2.2.4

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(g)	Materials:	3.3.1.1
(h)	Identification and marking:	3.3.3
(i)	Workmanship:	3.5
(j)	Preservation, packaging and packing:	5.1, 5.2
(k)	Marking for shipment and storage:	5.3

4.4.2 Caging tests.

4.4.2.1 Caged restriction. With the gyro caged by the motor shaft and wound as in 3.2.1.1, no total angular play about the outer gimbal axis shall exceed 1 degree when the gimbal is rotated alternately CW and CCW.

4.4.2.2 Caged indication. With the gyro caged and the motor fully wound, a resistance not exceeding 1.0 ohm, measured at 10 mAdc, shall appear between terminals 3 and 4 (Drawing 1563560). A resistance of not less than 10 megohms, measured at 100 Vdc, shall appear between terminals 2 and 4.

4.4.3 Actuation. Verify electrical characteristics and operation of the gyro actuator mechanism as follows.

4.4.3.1 Unfired actuator resistance. Measure the resistance between terminals 4 and 5 (Drawing 1563560) using 10 mAdc maximum current and with the gyro at 25 ±5°C. The resistance shall meet the requirements of 3.2.1.3.1.

4.4.3.2 Actuator no-fire current. Apply a current of 30 mAdc +0, -10 percent between terminals 4 and 5 (Drawing 1563560) for a period of not less than 5 minutes. Under this condition the actuator shall meet the requirements of 3.2.1.3.2.

4.4.3.3 Actuator firing current. Apply a current of 300 mAdc +10, -0 percent between terminals 4 and 5 (Drawing 1563560) for a period of 10 ms. Under this condition, the actuator shall release the spring motor in accordance with 3.2.1.3.3.

4.4.3.4 Uncaged indication. Monitor the resistance between terminals 2 and 4, and 3 and 4 (Drawing 1563560) from the onset of the current of 3.2.1.3.3 until the gimbal lock indication of 3.2.1.6.1.1. The gyro shall meet the requirements of 3.2.1.3.4 during this interval.

4.4.3.5 Runup time. With the gyro actuated in accordance with 4.4.3.3, verify that the gyro meets the requirements of 3.2.1.4.

4.4.3.6 Running time. Verify that the gyro running time to gimbal lock meets the requirements of 3.2.1.5.

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4.4.4 Gimbal freedom.

4.4.4.1 Inner gimbal freedom measurement. Inner gimbal freedom shall be measured with the gyro rotated first CW then CCW about the inner gimbal axis at 60 ±1 deg/sec. All measurements shall comply with 3.2.1.6.1 and 3.2.1.6.1.1.

4.4.4.2 Inner gimbal lock indication. Monitor the resistance between terminals 1 and 4 (Drawing 1563560) and verify that the inner gimbal lock indication meets the requirements of 3.2.1.6.1.1.

4.4.4.3 Outer gimbal freedom. Verify that the outer gimbal freedom motion requirements of 3.2.1.6.2 are met.

4.4.5 Pickoff. Verify that the pickoff system meets the requirements of 3.2.1.7 and related functional parameters as follows.

4.4.5.1 Excitation resistance. Under the conditions specified in 3.2.1.7.1, measure the resistance between terminals 6 and 7 (Drawing 1563560) with terminals 8 and 9 unconnected and the gyro at 25 ±5°C. The measured resistance shall meet the requirements of 3.2.1.7.1.

4.4.5.2 Resolution. Verify that the gyro meets the resolution requirements of 3.2.1.7.2.

4.4.5.3 Wiper contact. Verify that during the test of 4.4.6 the gyro meets the wiper contact requirements of 3.2.1.7.3. Noise voltage measurements for verification of compliance with 3.2.1.7.3, wiper contact, shall be via a lowpass filter with a 50-Hz cutoff frequency.

4.4.5.4 Sine and cosine approximation measurement. With pickoff excitation and wiper loading as in 3.2.1.7.2, the gyro shall meet the requirements of 3.2.1.7.4.

4.4.5.5 Uncaging offset. Verify that the uncaging offset requirements of 3.2.1.7.5 are met.

4.4.6 Drift. Gyro drift with roll shall meet the requirements of 3.2.1.8 when tested as follows:

4.4.6.1 Fast roll. Accelerate the gyro case to 260 ±10 deg/sec rotation in either direction about the outer gimbal axis alone (outer gimbal axis horizontal ±10 degrees), and actuate as in 4.4.3.3. The drift measured after the running time ±5 seconds specified in 3.2.1.5 shall not exceed 20 degrees.

4.4.6.2 Slow roll. Repeat 4.4.6.1 with a gyro rotational speed of 75 ±5 deg/sec. The drift measured after the running time ±5 seconds specified in 3.2.1.5 shall not exceed 20 degrees.

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4.4.7 Electrical tests.

4.4.7.1 Dielectric withstanding voltage. Upon application of 200 Vrms, 60 Hz between the gyro terminal under test and gyro case in accordance with MIL-STD-202, Method 301, and with terminals 4 and 5 jumpered (Drawing 1563560), the gyro shall meet the requirements of 3.2.1.9.1.

4.4.7.2 Insulation resistance. With terminals 4 and 5 jumpered, (Drawing 1563560) measure the insulation resistance between electrically isolated interior circuits and the gyro case in accordance with MIL-STD-202, Method 302, Test Condition A. The gyro shall meet the requirements of 3.2.1.9.2.

4.4.7.3 Terminal strength. Verify that the terminals meet the requirements of 3.2.1.10.

4.4.7.4 Actuator insulation resistance. Measure the resistance between terminal 5 (Drawing 1563560) and the gyro case, at 10 mAdc maximum. The gyro shall meet the requirements of 3.2.1.9.3.

4.4.8 Leakage test. Leakage measurements shall be made in accordance with MIL-STD-202, Method 112B. The fine leak test of 4.4.8.1 shall not be considered successful unless followed by the successful completion of 4.4.8.2.

4.4.8.1 Fine leak test. Fine leak test shall be performed in accordance with MIL-STD-202 Method 112B, Procedure IV, Condition C. Fine leak test is for preproduction and lot acceptance sample, not 100 percent individual acceptance test.

4.4.8.2 Gross leak test. Gross leak test shall be performed in accordance with MIL-STD-202, Method 112B, Condition B, and water may be substituted for silicone oil. A vacuum may be simulated by increasing the fluid temperature so the volume of gas sealed in the room temperature gyro increases in pressure as the trapped gases expand due to the heating of the warmed fluid (silicone oil or water). The appropriate temperature versus gyro gas volume shall be calculated or measured and justified prior to the use of the hot water alternate method. This bubble test must be performed to verify the fine leak test and is also performed as the 100 percent acceptance leak test.

4.4.9 Environmental tests.

4.4.9.1 Nonoperating environments. The gyro shall meet the nonoperating environmental requirements of 3.2.5.1 when tested as specified herein.

4.4.9.1.1 Temperature cycle. Gyro temperature shall be raised within 1 hour to +57 ±5°C, held at that high level for 4 hours minimum, lowered within 2 hours to -43 ±5°C, held at that low level for 4 hours minimum, and returned within 1 hour to room ambient. This constitutes the first of two such temperature cycles that shall be carried out. The vibration of 3.2.5.1.3

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shall be applied during the high and again during the low level dwell of the first temperature cycle, and the vibration of 3.2.5.1.3 similarly during the second temperature cycle.

4.4.9.1.2 Sine vibration. Sine vibration frequency shall be swept at logarithmic rate from 5 to 500 to 5 Hz in 15 minutes. Vibration level at the gyro attachment points, within ±10 percent of acceleration or amplitude, shall be ±1.3 g peak, 5 to -26 Hz; 0.036 inch double amplitude, 26 to 52 Hz; and ±5 g peak, 52 to 500 Hz. Time per application shall be four frequency sweeps (1 hour total) minimum per axis in each of three axes (see Figure 1).

4.4.9.1.3 Random vibration. Random vibration power spectral density at the gyro attachment points shall be 0.0250 ±0.0025 g^2/Hz from 20 to 2,000 Hz with 6 to 12 dB per octave rolloff below 20 and above 2,000 Hz. Time per application shall be 1 hour minimum per axis in each of three axes (See Figure 1).

4.4.9.1.4 Temperature shock and altitude. The gyro shall be held 1 hour minimum in a hot chamber at +71 ±5°C and ambient barometric pressure, transferred within 5 minutes maximum to and held for 1 hour minimum in a cold chamber at -43 ±5°C and 50,000-foot minimum barometric altitude, and returned within 5 minutes maximum to the hot chamber. This constitutes the first of six such cycles that shall be carried out. At the end of the last cycle the gyro shall remain in the hot chamber 1 hour minimum before being cooled to room temperature.

4.4.9.1.5 Handling shock. Handling shock shall be half-sine wave shocks of 20-ms minimum duration at 25 g minimum peak amplitude, applied three times per direction in each direction along each of three axes (see Figure 1) with the gyro at +57 ±5°C, and reapplied similarly with the gyro at -43 ±5°C, for an overall total of 36 shocks.

4.4.9.1.6 Humidity. The gyro shall be tested for moisture resistance in accordance with MIL-STD-202, Method 106, with vibration omitted.

4.4.9.2 Operating environments.

4.4.9.2.1 Temperatures. Actuate the gyro in accordance with 4.4.3.3 with the gyro at a temperature of -43 and +57 ±5°C, respectively. The gyro shall meet the requirements of 3.2.5.2.1 under each condition.

4.4.9.2.2 Acceleration. The gyro shall operate as specified herein when exposed to 25 ±3 g acceleration applied along the A-B axis only (see Figure 1), in the direction A to B, for 5 seconds minimum starting at the onset of the voltage of 4.4.3.3.

4.4.10 Data/certification inspection. Objective manufacturing data and contractor certifications shall be examined during collective inspection of a

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preproduction or periodic sample or preproduction lot of gyros for compliance with the following:

Item	Requirement paragraph
(a) Filling	3.2.2.3.2
(b) Materials, process and parts	3.3.1
(c) Cleanliness	3.3.8.1

5. PACKAGING

5.1 Preservation and packaging. Unless otherwise specified in the contract (see 6.2), preservation and packaging shall be in accordance with 5.1.2.

5.1.1 Level A. Gyros shall be cleaned by Process C-1 and individually protected and unit-packaged in accordance with Method IIb of MIL-P-116, without the use of contact preservatives. One gyro only shall be included in a unit package. Unless otherwise specified herein, the unit package shall be further packaged in intermediate containers conforming to PPP-B-566, Class 1 of PPP-B-636, PPP-B-665, or PPP-B-676. The gross weight of the intermediate container shall not exceed 100 pounds.

5.1.2 Level C. Preservation, packaging, and packing shall be in accordance with the contractor's commercial practices to afford protection from damage during shipment, within the scope of this specification, from the supply source to the first receiving activity.

5.2 Packing. Unless otherwise specified in the contract (see 6.2), packing shall be in accordance with 5.2.3.

5.2.1 Level A. Gyros packaged as specified herein shall be packed in overseas-type wirebound wood, wood-cleated fiberboard, wood-cleated plywood, nailed wood, or fiber (Class 2) boxes conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, or PPP-B-636, respectively. Shipping containers shall have case liners conforming to MIL-L-10547; the case liners shall be closed and sealed in accordance with the appendix thereto. Case liners for boxes conforming to PPP-B-636 will not be required provided the center and edge seams and joints are sealed with tape, at least 1-1/2 inches wide, conforming to PPP-T-76. Box closures and strapping shall be as specified in the applicable box specification or appendix thereto. Fiber boxes conforming to PPP-B-636 may be banded with tape conforming to Type IV of PPP-T-97 and appendix thereto in lieu of steel straps. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

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5.2.2 Level C. Gyros packaged as specified herein shall be packed in containers of the type, size, and kind commonly used for this purpose, in a manner that will ensure acceptance by common carrier and safe delivery at destination. Shipping containers shall be in compliance with the Uniform Freight Classification Ratings, Rules, and Regulations or the ratings, rules, and regulations for other carriers, as applicable to the mode of transportation.

5.3 Marking for shipment and storage. Unless the contract specifies otherwise (see 6.2), unit packages, intermediate packages, and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The gyro specified is intended for use in the guidance system of an air-launched missile.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and of the applicable drawing (see 2.1).

(b) When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of 14 gyros as specified in 3.4. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests and approval of the first article.

(c) Responsibility for inspection and inspection facilities, if other than as in 4.1.

(d) Level of preservation and packaging, if different than 5.1.

(e) Level of packing, if other than as in 5.2.

(f) Marking for shipment and storage, if different than 5.3.

Preparing activity:
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DOCUMENT IDENTIFIER AND TITLE
MIL-G-85117(AS) GYRO, STORED-ENERGY, FREE, ROLL POSITION INDICATING

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
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